Energy & Buildings 223 (2020) 110156

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enb

Deep learning and transfer learning models of energy consumption forecasting for a building with poor information data

Yuan Gao [a], Yingjun Ruan [a,*], Chengkuan Fang [a], Shuai Yin [b]

[a] School of Mechanical Engineering, Tongji University, Shanghai 200092, China
[b] Ministry of Housing and Urban-Rural Development Technology and Industrialization Development Center, 100044 Beijing, China

ARTICLE INFO

Article history:
Received 10 August 2019
Revised 7 May 2020
Accepted 17 May 2020
Available online 3 June 2020

Keywords:
Building energy forecasting
Transfer learning
Deep learning
Energy consumption

ABSTRACT

Precise prediction of energy consumption in buildings could significantly optimize strategies for operating building equipment and release the energy savings potential of buildings. With advances in computer science and smart meters, data-driven energy forecasting models, particularly deep learning models, are becoming increasingly popular and can achieve good prediction accuracy. However, these models require a multitude of historical data from predicted buildings for training, which are difficult to acquire for newly constructed buildings or buildings with newly established measurement equipment. In order to obtain satisfactory prediction accuracy under such poor information state, this paper proposes two deep learning models, which are a sequence-to-sequence (seq2seq) model and a two dimensional (2D) convolutional neural network (CNN) with an attention layer, and transfer learning framework to improve prediction accuracy for a target building. A case study of three office buildings is discussed to demonstrate the proposed method and models. Compared with the results of a long short-term memory (LSTM) network with poor information state, the seq2seq model improved forecast accuracy for a building with a small quantity of data by 19.69 percentage points in mean absolute percentage error (MAPE), and the 2D CNN model by 20.54 percentage points, on average.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

Globally, building energy consumption currently accounts for about 40% of total final energy consumption, making the optimization of building operational performance crucial in today's world [1]. The building energy consumption of China, for example, which has become the second largest energy consumer after the United States [2], accounts for about 30% of the country's total energy consumption. Therefore, the energy consumption of buildings has considerable energy savings potential. This requires the precise prediction of building energy consumption because the accuracy of the forecast will directly affect the control strategies of a building's energy equipment and thus affect the building's potential energy savings. For instance, the reduction of peak electricity demand, which can be classified a strategy of Demand Side Management, will definitely gain economic and environment achievements [3], since Peak demand is a major factor in the growth of investments in network infrastructure [4]. Therefore, an accurate building consumption prediction will benefit both electricity network and end users in demand management [5], which can further Reduce the operational risk of network [6].

Approaches for the prediction of building energy consumption can be divided into two types: physics models and data-driven methods [7]. Physics models of building energy consumption take into account parameters related to the characteristics of the building itself and obtain building energy data through a clear thermodynamic mechanism. Heiple and Sailor [8] divided public buildings into 11 types to construct 22 physics models for prototypical public buildings and forecast their energy consumption. However, the construction of a physics model for a building requires a large number of parameters (Building thermal physical parameters, energy equipment parameters and so on) and detailed system settings (Personnel activity schedule, air conditioning zoning and so on) related to the operation of the building, which are difficult to

Abbreviations: ANN, Artificial neural network; CNN, Convolutional neural network; CV-RMSE, Coefficient of variation of the root-mean-square error; LSTM, Long short-term memory; MAE, Mean absolute error; MAPE, Mean absolute percentage error; RMSE, Root-mean-square error; RNN, Recurrent neural network; seq2seq, sequence to sequence; NLP, Natural language processing; SVR, Support vector regression; SVM, Support vector machine, 2D, Two dimensional; MLP, Multilayer layer perceptron neural network; MARS, Multivariate adaptive regression splines; ELM, Extreme learning machine; SI, swarm intelligence.

* Corresponding author.
E-mail address: ruanyj@tongji.edu.cn (Y. Ruan).

https://doi.org/10.1016/j.enbuild.2020.110156
0378-7788/© 2020 Elsevier B.V. All rights reserved.

obtain. Besides, when using physics model for energy forecasting, model calibration is an essential and significant step to guarantee the accuracy of building models [6,9]. However, model calibration requires a lot of measurement data, which is hard to obtain. Therefore, it is difficult for physics models to achieve ideal forecasting accuracy.

Data-driven models rely on historical data to predict energy consumption [7]. Previous studies have shown that different data-driven models are effective in the prediction of various building energy consumption, including heating consumption, cooling consumption, electricity consumption and hot water usage. For instance, Yan et al. [10] used multilayer layer perceptron neural network (MLP) and the support vector regression (SVR) to predict short-term heating consumption; Roy et al. [11] applied a hybrid model of MARS (Multivariate Adaptive Regression Splines) and ELM (Extreme Learning Machine) to forecasting heating load. As for cooling consumption, Ahmad and Chen [12] used six data mining models to predict cooling load. Those six models are - tree bagger, Gaussian process regression, multiple linear regression, bagged tree, boosted tree and neural network. In 2018, Huang et al. [13] employ a Bayesian Network model to forecast the cooling consumption in commercial buildings. In the field of electricity consumption forecasting, Mat Daut et al. [14] conducted a detailed review of the building's electrical energy consumption prediction. The result of literature review shows that the hybrid of SVM and SI (swarm intelligence) methods has indeed presented superior performance for forecasting building electrical energy consumption. Leung et al. [15] used Artificial neural network (ANN) to predict electricity demand for cooling systems in buildings. In addition, Pang and O'Neill [16] performed a large-scale Monte Carlo simulation to simulate the hot water usage in hotels. In recent years, with the continuous improvement of the dimension of building energy consumption data, deep learning, as a method that can well process high-dimensional data, has been increasingly applied in the field of building energy consumption prediction [17,18]. Deep learning models can also be used as a separate feature extractor [19]. Fan et al. [20] compared the performance of different types of deep learning models, including convolutional neural networks and recurrent neural networks, on short-term load forecasting; Guo et al. [21] argued that a simple dense deep learning model can obtain better accuracy than the random forest or the gradient boosting model. However, according to these studies, all data-driven models require a multitude of historical data for their construction, a requirement that is difficult to meet in some new buildings or buildings with a newly constructed energy monitoring platform. Given that having historical energy consumption data for buildings is critical for the creation of accurate building energy consumption predictions [22], if a monitoring platform cannot supply the large quantities of historical data needed for data-driven models, it is hard for accurate forecasts to be obtained.

In light of these studies and the current problems with physics models and data-driven models mentioned above, the idea is to use transfer learning to address the problem of insufficient historical data when using data-driven models to predict energy consumption. Transfer learning is applied in solving real-world problems in a number of fields, including natural language processing [23,24], face verification [25], and visual tracking [26]. In the field of energy and buildings, Grubinger et al. [27] used transfer component analysis and online transfer learning to forecast temperature in residential buildings, and Mocanu et al. [28] used a deep belief network with reinforcement learning to build an online cross-building transfer learning model. Those two papers focused on online transfer learning. With regard to off-line transfer learning, Ribeiro et al. [29] proposed a novel method, Hephaestus, for data pre-processing and domain adaptation in order to use transfer learning via traditional machine learning support vector regression (SVR). Ribeiro's research team used four schools in Canada to verify their method. Each school used one year of data for calculating the factors of the Hephaestus method and another two years of data for training the model. The results showed that the mean absolute percentage error (MAPE) was about 10%, the best result achieved in the case study. Although Ribeiro et al.'s research is instructive, their study used an entire year of real data to calculate the trend factors for the Hephaestus method, which is a luxury given the status of energy consumption information in some buildings. In addition, Hephaestus involves complex pre-processing for transfer learning, and the prediction process of SVR model is not time dependent. Time dependent implies that the same features at two different time steps will lead to different output [30]. In other word, in SVR model, the output in the previous time step can barely affect the model output in current time step because there is no internal temporal method in SVR [31]. In order to process time series data, SVR needs external temporal methods, like add some temporal attributes in feature vector. Therefore, Ribeiro used four temporal attributes in the model, which will undoubtedly increase the feature dimensions of the whole model.

In the current research on deep learning, convolutional neural network (CNN) and long short-term memory (LSTM) network are widely used in the field of deep learning, especially in computer vision and natural language processing (NLP) [32]. The network structure of LSTM is particularly suitable for processing time series data and is widely used in the field of NLP. Since building energy consumption data is a kind of time-series data, this paper proposes LSTM to build transfer learning model. Due to the good performance of CNN model in feature extraction, especially image feature extraction in computer vision, this paper uses CNN model to extract features from data and construct high-dimensional features. Be that as it may, those two models still have some drawbacks, for instance, LSTM's structure is relatively complex and has many parameters, so the debugging of the model needs to prevent overfitting and underfitting. Besides, LSTM could not store information in a long period [32]. As for CNN, the overall structure of CNN model is relatively stable, and the tuning time may be long. However, these problems can be solved in practical applications, for instance, adding regularization mechanism to prevent overfitting.

Finally, to address the limitations of the studies discussed above and take advantage of CNN network and LSTM network, this paper proposes two transfer deep learning models: (1) a sequence-to-sequence (seq2seq) model, which uses LSTM as encoder and decoder, and (2) a 2D convolutional neural network (CNN) model with an attention layer to improve forecasting accuracy by transferring knowledge learned from similar buildings. It is worthwhile to denote that the attention layer was only combined with CNN model in this paper. The objective of our study was to determine whether these two models could improve prediction accuracy over a conventional model.

The paper is organized as follows: Section 2 introduces related concepts concerning transfer learning, the seq2seq model, and the attention layer. Section 3 illustrates the methods and the specific network used in this paper. Section 4 presents a case study of three office buildings using the model proposed in detail. Furthermore, Section 5 introduces the results and discussion part. Finally, Section 6 delivers our conclusions.

2. Background

Considering that transfer learning is rarely used in the field of building energy consumption prediction, it is necessary to clarify some basic concepts and background knowledge, this section introduces the concepts of transfer learning, the sequence-to-sequence (seq2seq) model, and the attention layer.

2.1. Transfer learning and sequential transfer learning

Transfer learning is a promising concept by which prior knowledge from one domain and task can be applied to another domain and task. This concept is inspired by humans, who have an innate ability to learn everything from scratch. Humans transfer and apply knowledge to process a wide variety of tasks using knowledge they have gained in the past. From a computer science perspective, such transfer is a framework, which can take advantage of knowledge and information from other source domains to improve performance for a target source. This subsection first introduces some basic definitions and then provides a specific definition for transfer learning. The definitions are as follows.

Definition 1 (Domain) [24,33,34]. A domain is denoted by $D = \{X, P(X)\}$, where X represents the feature space, which means the space of all the feature vectors. In the field of this paper, the feature space usually including average temperature, average humidity, wind speed and other parameters. Each parameter builds one dimension in the feature space. P(X) represents the marginal probability distribution.

Definition 2 (Task) [24,33,34]. A task is denoted by $T = \{Y, f(\cdot)\}$, where Y represents the label space, which means the space of all the label vectors. In this paper, label space only includes the energy consumption of the building to be predicted. $f(\cdot)$ represents an objective predictive function. For example, in this paper, the $f(\cdot)$ refers to seq2seq () and 2D-CNN (), which are two transfer learning model proposed in this paper. The purpose of the objective predictive function is to predict the relationship between the features and the label, the more specific task in this paper is to predicting the building energy consumption.

Definition 3 (Transfer Learning) [24,34,35]. Given two different domains combined with different learning tasks—i.e., a source domain D_S and learning task T_S, and a target domain D_T and learning task T_T—the intent of transfer learning is to improve the performance of the target task in D_T by using the knowledge in D_S and T_S, where $D_S \neq D_T$ or $T_S \neq T_T$.

Fig. 1 depicts the difference between the traditional learning and the transfer learning. Part (a) of the figure shows the working process of traditional learning, in which different tasks require different domains to train the model. In the perspective of building energy forecasting, traditional learning uses one building's historical data to predict energy consumption of itself. Part (b) shows how transfer learning can synthesize different domains as a whole, thus enabling the model to acquire knowledge from the source domains and improve its performance on the target task. Considering the field of energy forecasting, transfer learning can acquire the knowledge from some buildings to improve model's performance on the similar type of buildings.

During a study on transfer learning, it is necessary to address three main research problems [34]: (1) what to transfer, (2) how to transfer, and (3) when to transfer. There are also three categories of transfer learning: (1) inductive transfer learning, in which the target task and the source task are different; (2) transductive transfer learning, in which the source domain is different from the target domain while the tasks are same; and (3) unsupervised transfer learning, in which the target task is different and the data are not labeled. This study used an inductive learning method.

Now, building on this definition of transfer learning, we can go on to provide the definition of sequential transfer learning. Since the building consumption data is always time series data and may be suitable for sequential transfer learning.

Definition 4 (Sequential Transfer Learning, also known as Model Transfer) [24,36]. A transfer learning scenario in which source and target tasks are different and the training is running in sequence.

Fig. 2 shows the difference between ordinary transfer learning and sequential transfer learning. In the figure, T_s and T_t denote different learning tasks from the source domain and the target domain respectively; t represents the timeline; and $[t_1, t_2]$ and $[t_3, t_4]$ indicate the time intervals in each task, where $t_1 < t_2$ and $t_3 < t_4$. In ordinary transfer learning, $t_1 = t_3$ and $t_2 = t_4$; in other words, the two timing intervals are identical and occur at the same time. Under sequential transfer learning, $t_2 \leq t_3$. In the intermediate form, in which $t_3 < t_2$, the learning process is disjoint, and tasks are not completed according to a time sequence. It is more usual to use ordinary transfer learning, but when tasks are a time series or involve related work, sequential transfer learning may be useful [36]. However, more training samples are required for sequential transfer learning. It is sometimes useful to converge these transfer learning methods together into the intermediate form.

2.2. Sequence-to-sequence (seq2seq) model

Besides the transfer learning framework, the prediction model is also an important part. The sequence-to-sequence (seq2seq) model is also known as the encoder–decoder model. The original purpose of this model focused on problems of machine translation [37,38]. Therefore, the seq2seq model is a translation model, designed for translating one sequence into another. Fig. 3 shows the process of a simple seq2seq model, where x and o represents the input data and output data separately, h represents the hidden state of each time step in the encoder, s indicates the hidden state

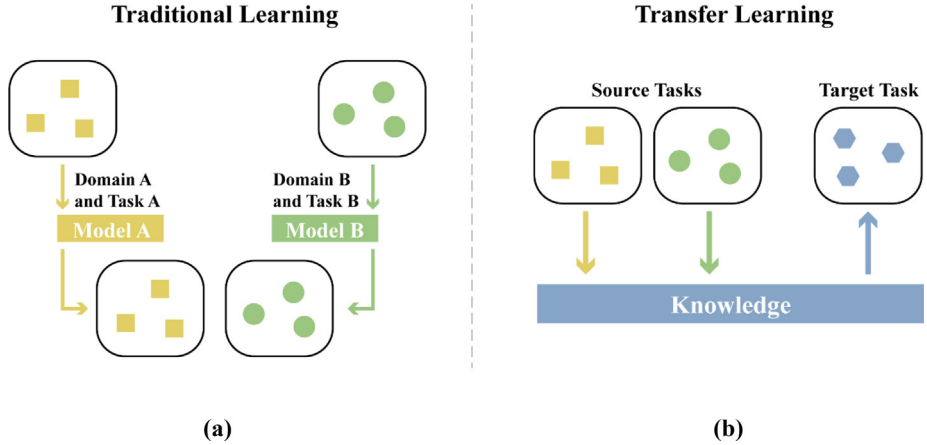


Fig. 1. The difference between traditional learning (a) and transfer learning (b).

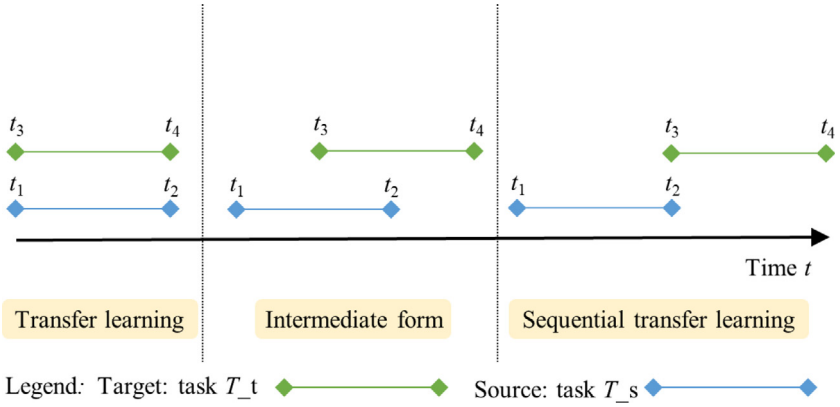


Fig. 2. The difference between ordinary transfer learning and sequential transfer learning.

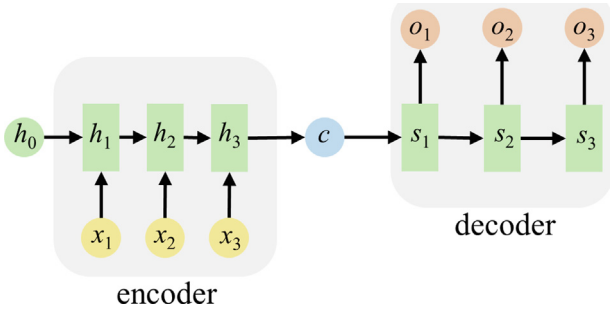


Fig. 3. Structure of simple seq2seq model.

in the decoder, and c denotes the context vector (which may be called the semantic vector or state vector.) In the field of natural language processing (NLP), the context vector refers the implication of natural language [32]. However, the context vector can be seen as the implication information of the *seq2seq* model. There are numerous ways to use the state vector c. In a simple seq2seq model, the context vector is set as the input to the decoder as shown in the Fig. 3. However, it is worthwhile note that we used the hidden state and the context vector together at each time step during the encoding procedure in this study. This is because this network structure does not limit the sequence length of input and output and can maintain good long-term memory. The structure is illustrated in Fig. 7 in detail.

2.3. Attention layer

It is commonly accepted that recurrent neural networks (RNNs) are suitable for solving time series tasks because of the special structure of RNN [39,40]. However, the calculation speed of an RNN is slow because it is not a parallel computational model. In addition, RNNs do not provide good performance on long-term prediction. In 2017, Google researchers [41] developed an entirely new layer, called the attention layer, to capture global trends and improve long-term prediction results. This equation gives the definition of the attention layer [41]:

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{1}$$

where $Q \in R^{n\times dk}$, $K \in R^{m\times dk}$, and $V \in R^{m\times dv}$ denote the query, key, and value, respectively. In the activation function softmax, three matrixes are multiplied, and the final result is an $n \times dv$ matrix. $\sqrt{d_k}$ plays a regulatory role, so that the inner product is not too large. This study further used the multi-head attention layer [41], which can be represented as follows:

$$\text{MultiHead}(Q,K,V) = \text{Concat}(head_1,\ldots,head_h)W^0 \tag{2}$$

where $head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right)$, $W_i^Q \in R^{d\ mod\ el\times dk}$, $W_i^K \in R^{d\ mod\ el\times dk}$, $W_i^V \in R^{d\ mod\ el\times dk}$, and $W^0 \in R^{hdv\times d\ mod\ el}$. The multi-head attention layer simply serves to project the queries, keys, and values h times without sharing weights.

Because the attention layer can capture global trends directly and compare two sequences at the same time, we use the attention layer with a convolutional neural network (CNN) to accomplish the transfer learning task.

3. Methods

In this section, we illustrate the entire process of transfer learning in detail using two deep learning models. Owing to the sparsity of good information on new buildings and buildings with new energy monitoring platforms, this model uses extra data from a similar building as a part of the training set in order to improve prediction performance and obtain a better prediction result. As the energy consumption data for the two buildings differ from each other, this task can be classified as inductive transfer learning. Fig. 4 provides an overview of this method, which consists of five parts: (1) data pre-processing, in which all training data and test data are processed into a relatively small scale to increase the calculation and convergence speed of the models (2) construction of the seq2seq model, in which an advanced seq2seq model is used to train the model and produce accurate prediction results (3) construction of the 2D CNN + Attention model, in which a conventional network and an attention layer are used to finish the task (4) pretraining and data augmentation, in which the training data are enhanced and the model calculates the context vector for the seq2seq model and (5) fine-tuning, in which the models transfer the knowledge from the source domain to the target domain and improve the prediction performance in the target domain. The following subsections describe these five parts of the method in turn.

3.1. Data pre-processing

Data pre-processing is a crucial step in both machine learning models and deep learning models and can improve the performance of the models considerably. Without pre-processing, the model will recognize those features that have relatively large values as contributing more to the model, the performance will be severely degraded, and the model may not converge. When the data pre-processing procedure is run, each feature in the model

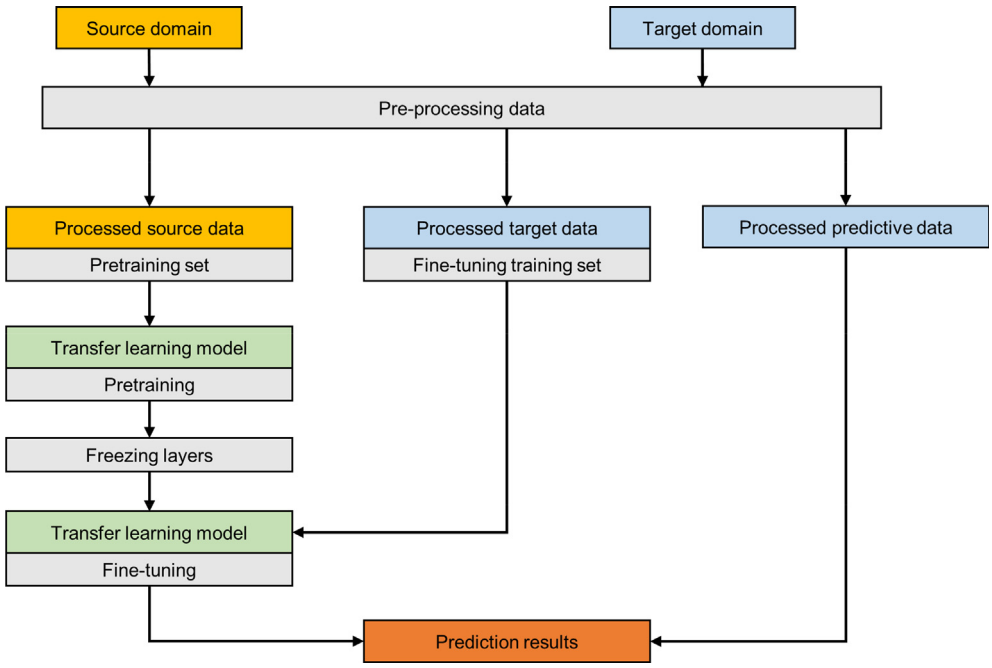


Fig. 4. Overview of transfer learning method.

can maintain a similar weight and make a similar contribution to the model. Thus, features from an entire range of scales can function as an integral part of the model. There are four main methods used for data pre-processing: min–max, z-score, log normalization, and one-hot code.

The min–max method is one of the most useful normalization approaches in machine learning and deep learning fields [42]. It is useful for data with a relatively lower variance and can maintain the stability of the data. In addition, it can maintain an entry of 0 in the sparsity matrix [42]. Min-max is represented by the equation

$$min - max = \frac{x - X_{\min}}{X_{\max} - X_{\min}} \tag{3}$$

where x is the value of the target feature; $X_{\min}$ and $X_{\max}$ are the minimum and maximum values, respectively, of this feature. This equation limits the value of the feature so that it lies between 0 and 1 before it is submitted to the model.

The z-score method, also known as standardization, is another widespread normalization method. The z-score can be represented by the following equation:

$$z = \frac{x - \mu}{\sigma} \tag{4}$$

where μ is the mean of all of the x values, and σ indicates their standard deviation. z is the result of standardizing x.

The z-score method does not limit the values to a specific range, and it can maintain the Gaussian distribution. However, an outlier can significantly influence the mean value of the x dataset. Therefore, it is sometimes reasonable to use the median value rather than the mean value. In addition, it may be difficult to obtain the proper normalization results from the test dataset using z-score because the mean value and the standard deviation of labels in the test dataset are unknown. Therefore, it is not reasonable for the labels in test dataset to use z-score. This problem can be solved by using log normalization for the labels [43]. The log normalization is presented by the following equation:

$$z = \log_a x \tag{5}$$

where x is the label, a is the bottom number of log function, and z is the result of log standardizing x.

Log normalization mitigates the adverse effects caused by poor numerical stability. Besides, log standardization also facilitates the estimation of physically meaningful parameters. The exponent of the estimated parameter limits the original parameter to a positive value. This limit is consistent with building energy consumption. During this paper, the bottom number in each experiment was set to 10 and selected z-score normalization method for each numeric feature.

One-hot coding is effective for scenarios in which a feature has categorical rather than continuous values. A one-hot code is a group of bits among which the only legal combinations of values are those in which there is a single high (1) bit and all of the other bits are low (0) [44]. This method can process attribute data and play a role in expanding features. One-hot coding was applied during this paper for each categorical feature.

3.2. Construction of encoder and decoder: long short-term memory (LSTM) and seq2seq model

3.2.1. LSTm

Traditional artificial neural networks (ANNs) can only address pairs of input and output individually and try to learn the relationship between inputs and outputs. However, some tasks need a better ability to process sequence information, meaning that previous output is related to the subsequent output. Fig. 5 shows the fundamental topology of a simple RNN, where X and o represent the input data and output data, respectively; h indicates the hidden layer; $\boldsymbol{U}$ denotes the weight matrix connecting the input layer and the hidden layer; $\boldsymbol{V}$ denotes the weight matrix between the hidden layer and the output layer; and $\boldsymbol{W}$ denotes the weight matrix between the two hidden layers in two different time steps. Accordingly, the output o_t in time step t will be influenced by the input X_{t-1} and the hidden layer state h in time step $t-1$. Therefore, the transfer process for the hidden layer state enables a simple RNN to have a memory of the previous sequence.

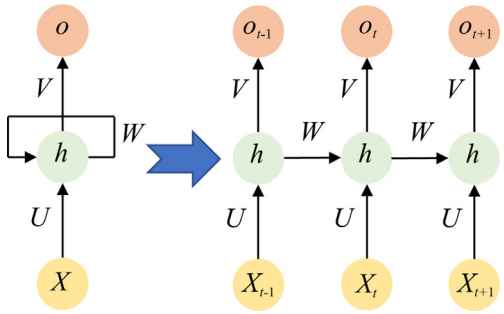


Fig. 5. Structure of a simple RNN.

Still, the simple RNN cannot convey information over a long time period. This is because after multiple iterations of the back-propagation algorithm, the gradient may become too small or too large [44]. Therefore, the simple RNN cannot remember long-term information. In order to remember long-term information, the LSTM neural network was designed [39]. Long short-term memory (LSTM) is an effective recurrent neural network (RNN). It can predict time series data and is more suitable for long-term time series prediction than simple RNNs. The LSTM neural network consists of LSTM units, which contain three special gates. The structure of an LSTM unit, called a cell, can be seen in Fig. 6.

In LSTM, the cell state is similar to a conveyor belt, which can be represented as a line from C_{t-1} to C_t. The cell state can run directly across the chain with only a few linear interactions. Thus, it will be easy to store long-term information in it. However, this single horizontal line is unable to add or delete information. In order to choose information, the LSTM units put information through structures called gates. There are three gates, which can select information through a sigmoid function and the dot product.

The first gate is called the forget gate, ft, i, which can provide a weight for forgetting the current state. The formulation of the forget gate in time step t can be represented as follows:

$$f_{t,i} = sigmoid\left(b_f(i) + \sum_j U_f(i,j)x_{t,j} + \sum_j W_f(i,j)h_{t-1,j}\right) \quad (6)$$

where the subscripts in $f_{t,i}$ indicate time t and cell i; $sigmoid(\cdot)$ represents the sigmoid activation function; b_f denotes the bias vector of the forget gate; U_f and W_f denote the input weight matrix and the regression weight matrix, respectively; x_t indicates the current input vector; and h_{t-1} denotes the current hidden layer vector.

The second gate is called the input gate, $in_{t,i}$, which can determine how much information can be accepted through the input data. The equation is represented as

$$in_{t,i} = sigmoid\left(b_{in}(i) + \sum_j U_{in}(i,j)x_{t,j} + \sum_j W_{in}(i,j)h_{t-1,j}\right) \quad (7)$$

where b_{in}, U_{in}, and W_{in} represent the bias vector and the input and output weight matrices of the input gate, respectively. Like the forget gate, the value of the input gate depends on the current input signal and the output of the hidden layer at the previous time step.

In addition, the LSTM cell will produce new cell memory $S_{t+1,i}$, which is provided as

$$S_{t+1,i} = f_{t,i} \cdot S_{t,i} + in_{t,i} \cdot C_{t,i} \quad (8)$$

$$C_{t,i} = \tanh\left(b(i) + \sum_j U(i,j)x_{t,j} + \sum_j W(i,j)h_{t-1,j}\right) \quad (9)$$

where $S_{t,i}$ and $C_{t,i}$ represent the old and new cell memories, respectively, and $\tanh(\cdot)$ is a kind of activation function.

The last gate, called the output gate $o_{t,i}$, calculates how much information can be seen as the output signal. The equation can be written as

$$o_{t,i} = \text{sigmoid}\left(b_o(i) + \sum_j U_o(i,j)x_{t,j} + \sum_j W_o(i,j)h_{t-1,j}\right) \quad (10)$$

where b_o, U_o, and W_o represent the bias vector and the input and output weight matrices of the input gate, respectively. As a result, the final LSTM output is defined as

$$h_{t,i} = o_{t,i} \cdot \tanh(S_{t,i}) \quad (11)$$

3.2.2. Seq2seq model

Based on the characteristics of the seq2seq model described in Section 2, this study used seq2seq as the main model framework for transfer learning. In order to improve the performance of the model and enable it to remember the long-term information, the study used two separate layers of LSTM as the encoder and the decoder. Fig. 7 shows the structure of this seq2seq model as used in the machine translation field. In the figure, $\boldsymbol{x}$ and $\boldsymbol{o}$ denote the input data and output results of the model, $\boldsymbol{c}$ denotes the context vector in the encoder, and the hidden states in the encoder and decoder are provided by $\boldsymbol{h_t}$ and $\boldsymbol{s_t}$, respectively, whose equations are as follows:

$$h_t = LSTM_{enc}(x_t, h_{t-1}) \quad (12)$$

$$s_t = LSTM_{dec}(o_{t-1}, s_{t-1}) \quad (13)$$

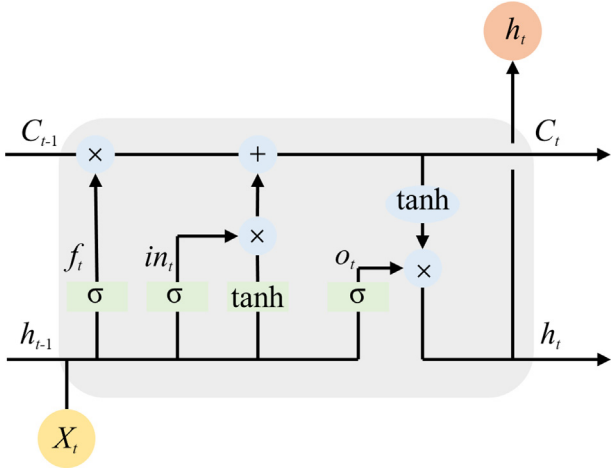


Fig. 6. Structure of LSTM cell.

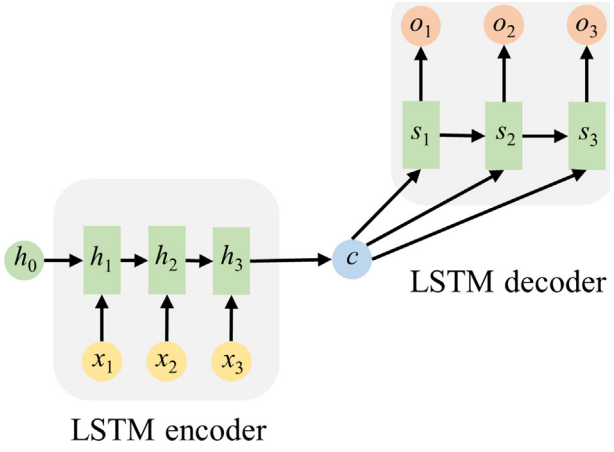


Fig. 7. Structure of advanced seq2seq model.

The context vector $\boldsymbol{c}$ is calculated by all of the hidden states in the encoder LSTM layer, and this context vector will be replicated and submitted to each decoder unit.

3.3. Construction of CNN with attention layer

With advances in measurement technology and computer science, we can now acquire building datasets with large numbers of dimensions, which can help us obtain a good prediction result for energy consumption in buildings. Thus, it is important to take advantage of such multidimensional data to build a prediction model with more advanced features. However, as the number of data dimensions rises, it becomes increasingly difficult to build high-level features manually. In order to obtain high-level features, this study used a 2D CNN for feature extraction and proposed a novel 2D CNN model with an attention layer as a framework for transfer learning and forecasting energy consumption.

In the field of computer vision, the 2D CNN is an effective tool for feature extraction [45]. Owing to the flexibility of the convolution kernel, it is easy to extract a multitude of features through feature maps. In order to use the 2D CNN to extract features, we reconstruct our one-dimensional data features as two-dimensional data features. Fig. 8 shows how this feature extraction process operates. For the sake of the 2D CNN, the original one-dimensional features are reshaped into two dimensions as the reshaped features in Fig. 8. In this case, it is the flexibility to apply different sizes and numbers of convolution kernels to extract high-level features. Such features could improve the model performance since the convolution kernels can be set as different size and on different locations. For instance, this paper uses the convolution kernels of 2*2 which is shown in Fig. 8. In addition, we can use padding [46] to avoid information loss.

Although the model will extract high-level features through the 2D CNN neural network, it is still important to consider the dependence of time series in the training and prediction process. Therefore, the training datasets should not be shuffled. The multi-head attention layer is used in this model to capture the feature of temporal dependency.

3.4. Pretraining and data augmentation

The concept of pretraining appeared when researchers found that an artificial neural network can be applied to specific target domains after training on different source domains. The researchers found that the training process on source domains improve the performance on target domains [47]. After that, it was realized that a model can be pretrained before being trained for specific problems. Pretraining is a key step in the 2D CNN with attention layer model and seq2seq transfer learning model especially because pretraining can obtain an intact context vector that will contain knowledge or useful properties from the source domain. Normally, there are two ways of pretraining: (1) pretraining for the extraction of features, by which a model will be trained on a similar domain in order to extract a sufficient number of high-level features, and (2) pretraining a model on a larger source domain and then freezing some top layers and fine-tuning those layers using target domain data.

In this study, the pretraining method was used to extract knowledge from the source domain data and transfer the knowledge to solve the tasks in the target domain.

In order to create a good pretrained model, it is important to obtain a large quantity of labeled training data. However, labeled data, especially building consumption data, are difficult to obtain because special equipment is required to take the measurements and a long period of time is needed to collect the data. Accordingly, data augmentation [48] has been proposed as a technology for increasing the quantity of data trained to improve a model's generalizability and robustness. However, owing to the special input data dimensionality of LSTM, which the data needs to be reshaped into the same long of timestep, and the special requirements of sequential transfer learning, this study proposed a method for enhancing the time series data in the LSTM and seq2seq model. Fig. 9 shows how this method works. And this method will only be applied on seq2seq model. In the figure, the raw training data are the original training data, which are in a sequence of length m, and the sample window denotes the sliding window for the sampling process and has length n, where $n < m$ and the length of the sample window will be the same during the process.

In this case, without data augmentation, the number of training samples will be $T_1 = m/n$. After data augmentation, however, the number of training samples will be $T_2 = m - n$. As long as $T_2 > T_1$, this technology will certainly increase the number of training samples. As n will normally be much less than m, the number of training data will definitely be increased. In addition, the length of sample window will be the same during the whole process. Considering the periodicity of office buildings, the time step length of data input of LSTM model in this paper is all 7, that is, m = 7. Furthermore, as the CNN model will consider temporal dependency using a multi-head attention layer, this data augmentation will only apply to the model associated with the LSTM layer. Considering that the number of training samples after data enhancement will far exceed the prediction target, we add regularizations in the LSTM layers and convolution layers of the model to prevent overfitting and enhance the generalizability of the model.

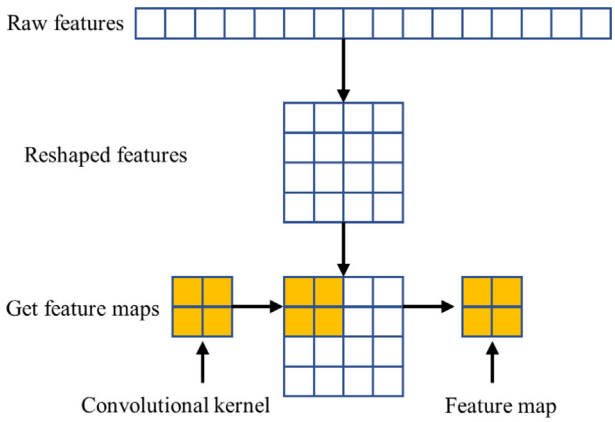


Fig. 8. Construction of feature maps using 2D CNN.

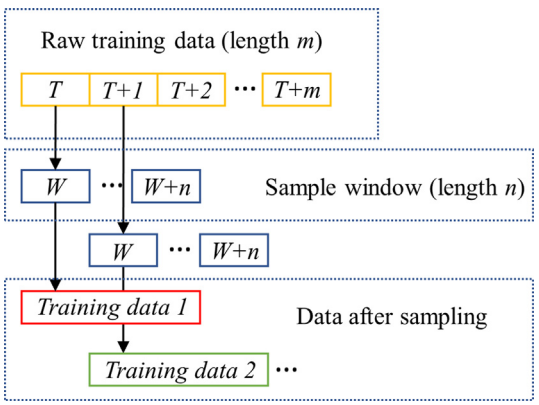


Fig. 9. Sampling method for data augmentation.

3.5. Fine-tuning

Fine-tuning trains a pretrained model on the target domain data and can update the knowledge and representations that were learned by the pretrained model [49], thereby enabling the results of pretraining to be taken advantage of. Fine-tuning is a crucial method in deep learning and transfer learning fields because of its feasibility and convenience. In order to meet the requirements of transfer learning and extract the knowledge from the source dataset, our seq2seq model and 2D-CNN model need a fine-tuning process.

According to previous research [50], the lower layer in the deep neural network always contains more general features. In contrast, the top layer may contain more special characteristics. Based on the characteristics of neural networks, there are four different scenarios we may meet and four corresponding categories. Fig. 10 depicts a coordinate system that categorizes the four scenarios and indicates the method for dealing with each type.

In quadrant 1 of Fig. 10, the target domain dataset is different from that of the source domain but is large in size. Therefore, because we have an abundance of data, we can train a new model from scratch instead of fine-tuning the pretrained model. The pretrained model can be used as the feature exactor. In quadrant 2, the dataset from the target domain is large in size and is similar to that of the source domain. In this case, although we may obtain a good result with any strategy, using the target data to fine-tune the pretrained model may be the most effective because the model can take advantage of the knowledge from the source domain, and the fine-tuning process will be quicker than training from scratch. Quadrant 3 illustrates a very difficult situation, in which the target dataset is small and has little similarity to the source domain. Therefore, we may consider fine-tuning only the lower layer(s), but this may not yield a satisfactory result because of the small dataset size. Therefore, we should also consider applying a data augmentation technique. Quadrant 4 depicts a small target dataset that is similar to the source domain dataset. Here, we could fine-tune just the top layer(s) of the model (usually a dense layer) to transfer the knowledge from the source domain.

In this study, when there was a seq2seq model in the transfer learning, the main purpose of pretraining was to train the context vector; this is because in the machine translation field, the context vector holds specific information about the knowledge and meaning of the entire sequence. After pretraining is complete, there is a fine-tuning process using a small quantity of similar data from the target set. In the quadrant 4 scenario of Fig. 10, before fine-tuning is begun, the encoder of the seq2seq model must be frozen in order to maintain the context vector from the source dataset. When there is a CNN with a multi-head-attention-layer model in a transfer learning scenario, the CNN layers and the attention layer of this model will be frozen, as shown in quadrant 4.

4. Case study

This section describes a case study in which we applied the seq2seq model and the 2D CNN model with an attention layer to improve the prediction performance for a building with little high-quality information available. The expectation was that with the help of data on other buildings that are similar, this transfer learning framework would be able to deliver a satisfactory prediction result. The goal for these two models was to obtain an error rate as low as that of the model that had been trained using two whole years of energy consumption data from the target domain.

4.1. The introduction of buildings

We used three similar government office buildings from Shandong province in China. There are three experiments described in this section, each treating a different building as the target building. The three buildings are located in two different cities. The building energy consumption data that were collected by the energy consumption platform include three main parts: (1) daily equipment electricity usage, (2) lighting, (3) consumption by the air conditioning system in summer (because the sites are located in northern China, the three buildings are centrally heated by the local government during the winter season). Therefore, all the data in this article are electricity consumption data. Fig. 11 provides a map showing the three buildings' geographical locations.

The data were collected and provided by the government. The three office buildings differ in the size of the area covered (Building A: 23679 m^2; Building B: 15520 m^2; Building C: 26768 m^2) but are located in the same province, and therefore, they share the same relative weather conditions and thus the same meteorological features. In addition, because all three buildings are government office buildings, they run an identical work schedule during the business day and share similar holidays throughout the year, following the local government.

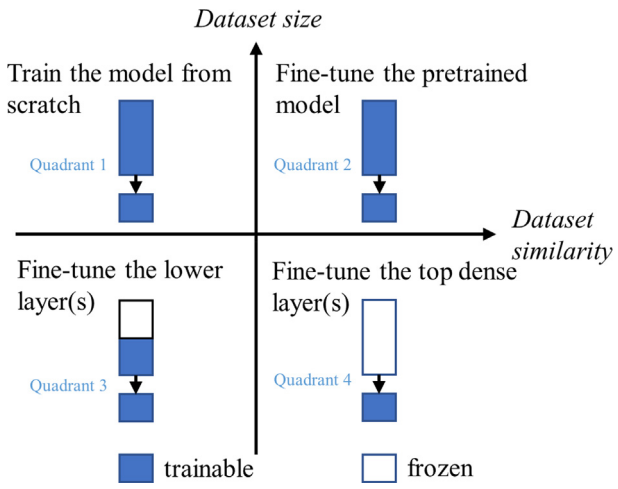


Fig. 10. Fine-tuning strategies.



Fig. 11. Geographical location map of the three buildings in the case study.

4.2. Datasets

The datasets contained three years' daily data from January 1, 2015, to December 31, 2017, for each building. Each dataset contained twelve features, including ten numeric features (mean temperature, maximum temperature, minimum temperature, mean humidity, maximum humidity, minimum humidity, dew point, mean atmospheric pressure, mean wind speed, maximum wind speed) and two categorical features (holiday, which is a binary feature denoting whether it is a holiday, month of the year) as well as the label energy consumption, which is collected by the government. Fig. 12 shows the frequency histogram of each building's energy consumption in 2016. The figure demonstrates that each building has a different histogram for the year. However, the kernel density estimates for each building have relatively similar distributions, which indicates that these three buildings are suitable for transfer learning because the knowledge extracted from one building may be usefully applied to other buildings. The subsections that follow discuss the evaluation of transfer learning in the seq2-seq model and the 2D CNN model with an attention layer and provide the case study results and discussion.

4.3. Experiment setup

The evaluation was performed on the results from three experiments, each of which included five models, to assess the effectiveness of the two transfer learning models. However, it is worth noting that all of the prediction processes used the ground truth value of the features. Therefore, the prediction results are ideal results because in the real world there is still some error in the prediction of meteorological parameters, and this part of the error is not considered in this study. Table 1 shows how the source domain and target domain were created for each experiment on transfer learning.

Table 1
Division of data into source domain and target domain.

Experiment	Target domain	Source domain
1	Data on building A from 2016 to 12 to 2017–12	Data on buildings B and C from 2015 to 2016
2	Data on building B from 2016 to 12 to 2017–12	Data on buildings A and C from 2015 to 2016
3	Data on building C from 2016 to 12 to 2017–12	Data on buildings A and B from 2015 to 2016

In the case study of this paper, the source field refers to the data set composed of all buildings except the target building, while the target field refers to the data set of the current buildings that are expected to be predicted. In order to better prove the robustness of the model, this paper will use building A, building B, and building C respectively as target buildings for transfer learning prediction. For example, when building A is the target building, the target domain is the data of building A, and the corresponding source domain is the data of building B and building C. At the same time, in order to meet the requirements of sequential transfer learning which can be seen in Fig. 2, the target domain data should lag behind the source domain data in time. That is, the data of source domain should be the data of 2015–2016, while the data of target domain should be the data of December 2016 and the whole year of 2017. It is worth noting that the data of December 2016 in the target domain is the data used for fine-tuning the forecasting of the target building in January 2017.

In order to assess the prediction results of the seq2seq model and the CNN model, we defined five models to compare:

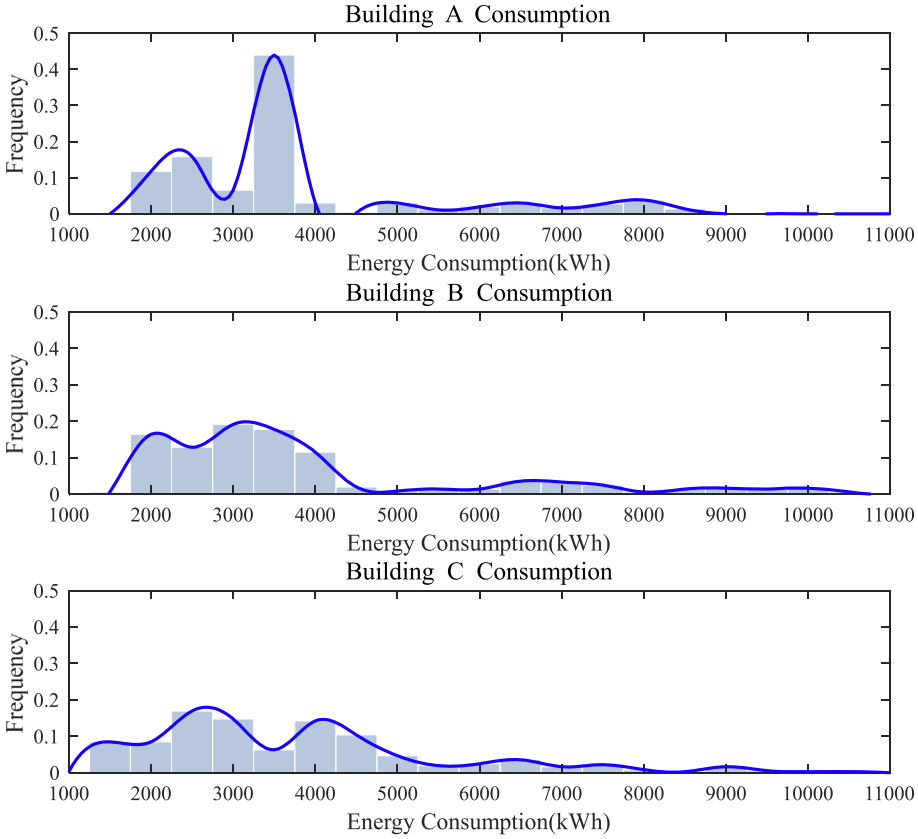


Fig. 12. Frequency histogram for each building in the case study.

- **M.1**: Model 1 used simple LSTM layers and a dense layer to build the model. This model used only one month of data from the target building for training. The purpose of this model was to illustrate the current condition of poor information and to set a baseline for evaluating the effectiveness of the other models.
- **M.2**: Model 2, like model 1, used simple LSTM layers and a dense layer to build the model. However, model 2 used two years of historical data from the target building for the training process. This model was used as the benchmark to evaluate the two transfer learning models.
- **M.3**: Model 3 used 2D CNN layers and a multi-head attention layer for building the transfer learning model. All of the data from the source domain were used for pretraining, and only one month was used as a fine-tuning dataset in the target domain for each prediction.
- **M.4**: Model 4 used two LSTM layers as individual encoder and decoder to build a seq2seq model. The training process settings were the same as those for model 3.
- **M.5**: Model 5 used simple LSTM layers and a dense layer for transfer learning. The pretraining and fine-tuning processes were identical to those of model 3 and model 4. The purpose of this model was to assess the necessity of applying models as complex as model 3 and model 4.

Table 2 briefly summarizes the information for the five models.

We used the Python 3.6 programming language and the Keras 2.2.4 deep learning framework to implement our model. Fig. 13 shows the architectures for all five models, designated as architectures 1, 2, and 3. Models 1, 2, and 5 applied architecture 1; model 3 used architecture 2, for which we set a 2*2 convolutional kernel in the 2D CNN layers; and model 4 used architecture 3. The 2D CNN layers used the regularization function that comes with Keras to prevent overfitting, and the LSTM layers used the dropout mechanism to prevent overfitting [51]. Furthermore, each model optimized the hyperparameters, such as the number of CNN layers, the dropout parameters, and the number of neural units during model training, to ensure the reliability of the prediction results.

The prediction process provided rolling forecasts for one year, for which the fine-tuning data were only a month ahead of the prediction target. Hence, there were twelve predictions to finish the transfer learning for each building in 2017. Table 3 shows the details of the rolling forecasts. However, as models 1 and 2 are not relevant to transfer learning, model 1 used only one month of data on the target building to predict the energy consumption of the following month without any pretraining process, and model 2 used the two years of historical data on the target building for pretraining.

Table 2
Summary of the five models that were compared.

Model ID	Specific model	With/Without transfer learning	Data used
M.1	LSTM model	Without transfer learning framework	One month's data from target building for each prediction
M.2	LSTM model	Without transfer learning framework	Two years' data from target building for each prediction
M.3	2D CNN model with attention layer	Using transfer learning framework	Two years' data from each building in source domain and one month's data from target building for each prediction
M.4	Seq2seq model	Using transfer learning framework	Two years' data from each building in source domain and one month's data from target building for each prediction
M.5	LSTM model	Using transfer learning framework	Two years' data from each building in source domain and one month's data from target building for each prediction

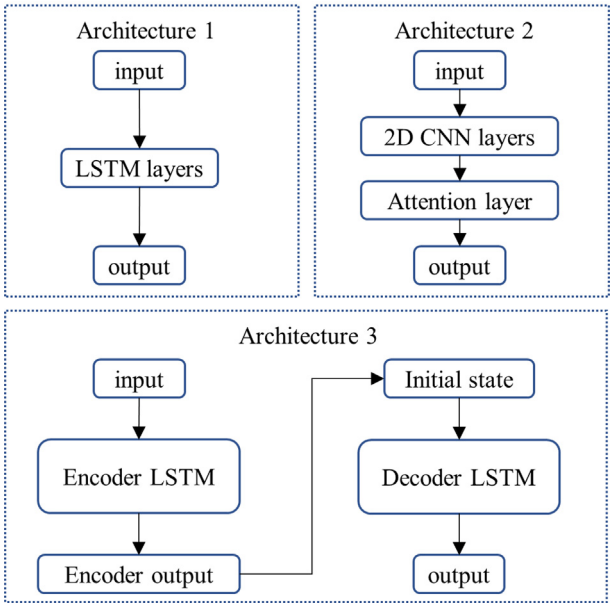


Fig. 13. Model architectures in this study.

In order to better describe the training data and test data in the five models, the following is an example to predict the energy consumption of Building A in January 2017.

According to Fig. 4, the model involved in transfer learning is divided into source data, target data and predictive data, while the model not involved in transfer learning is described by train data and test data. Due to space limitation, the description forms of M.1, M.2 and M.5 are shown in the Appendix (Tables 10–12), only M.3 model and M.4 model are shown here

Tables 4 and 5 show the data content, sampling rate, data role and data dimension contained in different data in M.3 and M.4. The specific content of features and label in the table can be seen in Section 4.2. In this paper, the time steps of input data of LSTM network are all 7. Due to the particularity of seq2seq's network structure (one layer of LSTM as encoder and one layer of LSTM as decoder), a special iterative process is needed for training and prediction. Table 6 details this iterative process with the example of predicting the energy consumption of building A in January 2017.

The expected results were that the accuracy of model 1 would be worse than that of models 2, 3, and 4 because model 1 does not transfer knowledge from similar buildings. In addition, model 2 was expected to obtain the best accuracy because it used the large target building historical dataset, and models 3 and 4 would be somewhat lower in accuracy than model 2 but would still obtain relatively good results compared with model 1. In model 5, which was included to evaluate the necessity of using more complex models, we also applied the transfer learning process; even so, it was considered an open possibility for the results of model 5 to be worse than those of models 3 and 4.

5. Results and discussion

5.1. Evaluation method

For this study, we selected mean absolute percentage error (MAPE) and the coefficient of variation of the root-mean-square

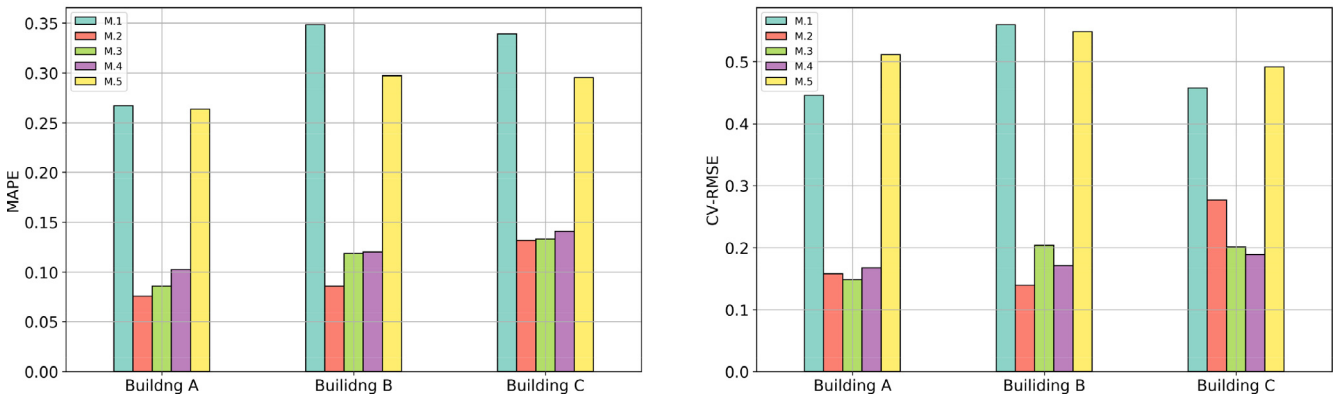


Fig. 14. Comparison of energy consumption prediction errors in each experiment.

Table 3
Rolling forecasting process.

Forecast order	Target predictive data	Fine-tuning data	Pretraining data
1st	The data for 2017–1 on the target building	The data for 2016–12 on the target building	Corresponding source domain data
2nd	The data for 2017–2 on the target building	The data for 2017–1 on the target building	Corresponding source domain data
3rd	The data for 2017–3 on the target building	The data for 2017–2 on the target building	Corresponding source domain data
⋮	⋮	⋮	⋮
11th	The data for 2017–11 on the target building	The data for 2017–10 on the target building	Corresponding source domain data
12th	The data for 2017–12 on the target building	The data for 2017–11 on the target building	Corresponding source domain data

Table 4
Data information of M.3.

	Data content	Sampling rate	Function
Source data	The data for 2015 to 2016 on the building B and building C (Features and Labels)	Daily	Pretraining
Target data	The data for 2016–12 on the building A (Features and Labels)	Daily	Fine-tuning
Predictive data	The data for 2017–1 on the target building (Features)	Daily	Final target month forecast

Table 5
Data information of M.4.

	Data content	Sampling rate	Function
Source data	The data for 2015 to 2016 on the building B and building C (Features and Labels), using iterative process	7 days	Pretraining
Target data	The data for 2016–12 on the building A (Features and Labels), using iterative process	7 days	Fine-tuning
Predictive data	The data for 2017–1 on the target building (Features), using iterative process	7 days	Final target month forecast

Table 6
The description of iterative process.

Prediction target	Input data	Note
Energy consumption of building A on January 1, 2017	Features and labels of building A (December 25, 2016 solstice December 31, 2016) & Features of building A (January 1, 2017)	All labels are raw data at this point
Energy consumption of building A on January 2, 2017	Features and labels of building A (December 26, 2016 solstice January 1, 2017) & Features of building A (January 2, 2017)	The label on January 1, 2017 is the predicted value at this point
⋮	⋮	⋮
Energy consumption of building A on January 31, 2017	Features and labels of building A (January 24, 2017 solstice January 30, 2017) & Features of building A (January 31, 2017)	All labels are predicted value at this point

error (CV-RMSE) to assess the prediction accuracy. Because the buildings in our dataset are differ in the size of the areas they cover, the value of energy consumption may not on the same level of magnitude. Therefore, it was not suitable to use the mean absolute error (MAE) or the root-mean-square error (RMSE) as evaluation metrics between different experiments. However, the mean absolute error (MAE) was calculated to compare the prediction accuracy within a single experiment. MAPE and CV-RMSE were calculated over the entire prediction timeline. In addition, MAE was calculated for each time step. MAPE, CV-RMSE, and MAE are calculated respectively as follows:

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\frac{|y_i - \hat{y}_i|}{y_i} \tag{14}$$

$$CV\text{ - }RMSE = \sqrt{\frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{N}} \Big/ \frac{\sum_{i=1}^{N} y_i}{N} \tag{15}$$

$$MAE = \frac{|y_i - \hat{y}_i|}{N} \tag{16}$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and N is the number of observations. Table 7 shows the results for the three experiments. For this table, the average MAPE and CV-RMSE values were calculated for each model and each experiment. In addition, in the appendix (Fig. 16–18), in each building in winter, summer and transition season, three months (January, July, and October) are selected as a line chart for reference, where 'gnd' refers to the true

Table 7
Evaluation metric results for model predictions for each office building.

Model ID	Building A		Building B		Building C	
	MAPE	CV-RMSE	MAPE	CV-RMSE	MAPE	CV-RMSE
M.1	0.2672	0.4457	0.3482	0.5592	0.3391	0.4577
M.2	0.0760	0.1578	0.0859	0.1394	0.1317	0.2767
M.3	0.0860	0.1482	0.1187	0.2038	0.1334	0.2009
M.4	0.1025	0.1675	0.1201	0.1707	0.1411	0.1889
M.5	0.2636	0.5112	0.2971	0.5485	0.2952	0.4913

measured energy demand. Fig. 14 charts the values given in Table 7 in order to provide a more direct perspective.

5.2. Comparison of M.3 and M.4 with M.1

From the evaluation metric results for the experiments for each building as shown in Fig. 14 and Table 7, it can be seen that M.1 obtained the worst MAPE for all the buildings and the worse CV-RMSE for building B; M.1 also got the second worst CV-RMSE for building A and building C. This is reasonable because this model used only one month of data from the target domain to predict the energy consumption for the following month. This demonstrates the importance of using additional data on similar buildings to increase the forecast accuracy. According to Table 7, comparing the results of M.1, M.3 and M.4 on each building, building B showed the best improvement over M.1; for M.3 of this building, MAPE was reduced by 22.95 percentage points and CV-RMSE by 35.54 percentage points; for M.4 of this building, MAPE was reduced by 22.81 percentage points and CV-RMSE by 38.85 percentage points; M.4 on building A resulted in the lowest improvement of MAPE among all the experiments, though it is still considerable, with MAPE reduced by 16.47 percentage points. Across all buildings, M.3 improved forecasting accuracy over that of M.1 by 20.54 MAPE percentage points on average and by 30.32 CV-RMSE percentage points on average; M.4 improved forecasting accuracy over that of M.1 by 19.69 MAPE percentage points on average and by 31.18 CV-RMSE percentage points on average. Therefore, the two transfer learning models, M.3 and M.4 were able to improve the prediction accuracy.

5.3. Comparison of M.3 and M.4 with M.5

M.5 attained the second worst MAPE in each experiment and the worst CV-RMSE for building A and building C, which indicates the necessity of applying a more complex model because simple LSTM network can hardly perform well under transfer learning framework. Using the results of M.5 as a baseline, M.3 on building B obtained the best improvement with MAPE reduced by 17.84 percentage points; M.4 on building B obtained the best improvement with CV-RMSE by 37.78 percentage points. The lowest improvement on MAPE came from model M.4 on building C, with MAPE reduced by 15.41 percentage points, and the lowest improvement in CV-RMSE came from model M.3 on building C, where CV-RMSE was reduced by 29.04 percentage points. Besides, the results of M.5 on each building are similar to M.1, which are terrible comparing with M.3 and M.4. Therefore, it was significantly worthwhile in these cases to apply the more complex models, M.3 and M.4, as these models can improve the accuracy over that of M.5.

5.4. Comparison of M.3 and M.4 with M.2

As for M.2, which used two years of historical data on the target building in each experiment and was included as a benchmark, the results were as expected on each building, for which M.2 achieved the best accuracy in MAPE. The results of M.3 and M.4 were very close to M.2, which could demonstrate the effectiveness of two transfer learning models. In terms of CV-RMSE, however, M.3 performed better than M.2 on building A, and M.3 and M.4 were better than M.2 on building C. This may be due to the fact that M.2 and M.3 use the data of four entire years from the two buildings in source domain whereas M.2 uses only two years from the target building, enhancing the generalizability of M.3 and M.4 because of the greater number of training samples. However, an understanding of the specific reasons will require further research.

5.5. Comparison of M.3 with M.4

From the results for M.3 and M.4 given in Fig. 14 and Table 7, it is difficult to determine which model is better. Thus, this study calculated the MAE for each model by using forecasting results and true values for each day of 2017 to figure out if there is any extremely prediction result during whole year, and thus assess the performance of M.3 and M.4. The results are illustrated by box plots in Fig. 15. The purpose of these figures is to illustrate the results for each experiment in detail, particularly for M.3 and M.4. From the first part of Fig. 15, showing the results for building A, it can be seen that the maximum outlier result for M.4 is more extreme than that for M.3 and for the mean value of MAE, the results of M.3 is better than M.4 on building A under this perspective. The overall MAE distribution of the two models is also relatively close. From the second part of Fig. 15, which displays the results for building B, it can be seen that the maximum outlier result for M.3 is more extreme than that for M.4, which may lead the CV-RMSE for M.3 is 3.31 percentage points larger than that for M.4. From the third part of Fig. 15, showing building C, it is worth noting that M.2 has a more extreme maximum outlier than M.3 and M.4. Therefore, in terms of CV-RMSE for building C, M.3 and M.4 performed better than M.2 (the CV-RMSE for M.3 and M.4 are 7.58 and 8.78 percentage points lower than that for M.2).

5.6. Use and limitations of the transfer learning model

In order to further verify the use and limitations of the transfer learning model, this paper takes building B as the target building and constructs three supplementary experiments for the seq2seq model and 2D-CNN model by changing the data size and data similarity degree of the source domain. The supplementary Experiments only change the source domain data in the transfer learning, and the target domain data and prediction method are consistent with M.3 (2D-CNN) and M.4 (seq2seq) in Experiment 2, which can be seen in Section 4.3. The details are shown in Table 8.

Similarly, MAPE and CV-RMSE were used in supplementary experiments to evaluate the accuracy of prediction results. Table 9 shows the results for the three experiments. In addition, the results of 2D-CNN and Seq2seq from the original experiment 2 are also included in the table for readers to compare.

The prediction accuracy of S.1 model is not very different from that of the original experiment. Considering the results of MAPE, S.1 experiment was 2.3 percentage points higher than the original

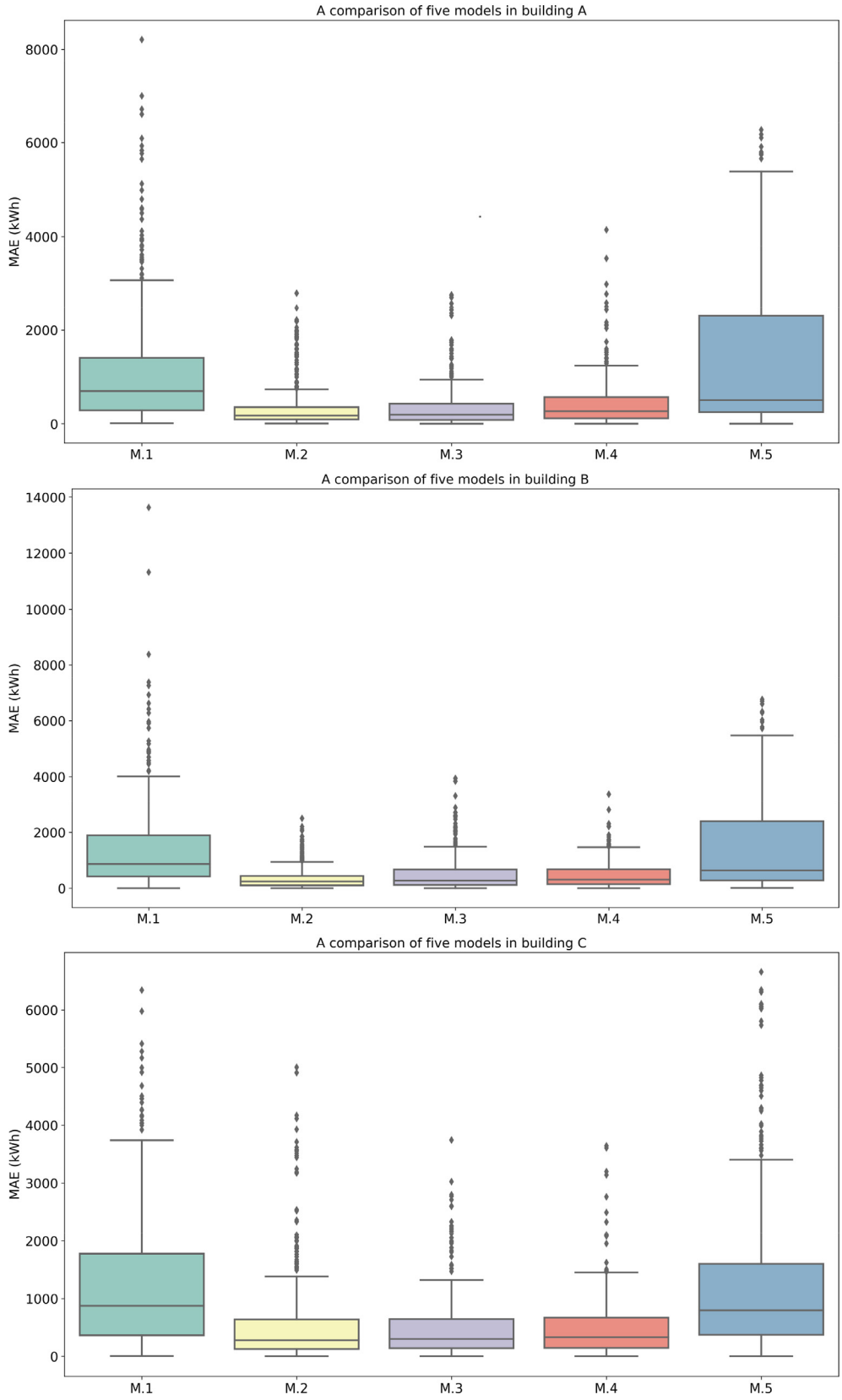


Fig. 15. Daily MAEs of different models in 2017.

experiment on the 2D-CNN model and 1.96 percentage points higher on the Seq2seq model. Besides, the difference between CV-RMSE is also not very large. This indicates that the transfer learning model proposed in this paper can also achieve relatively good prediction accuracy when only one year of data is available in the source domain.

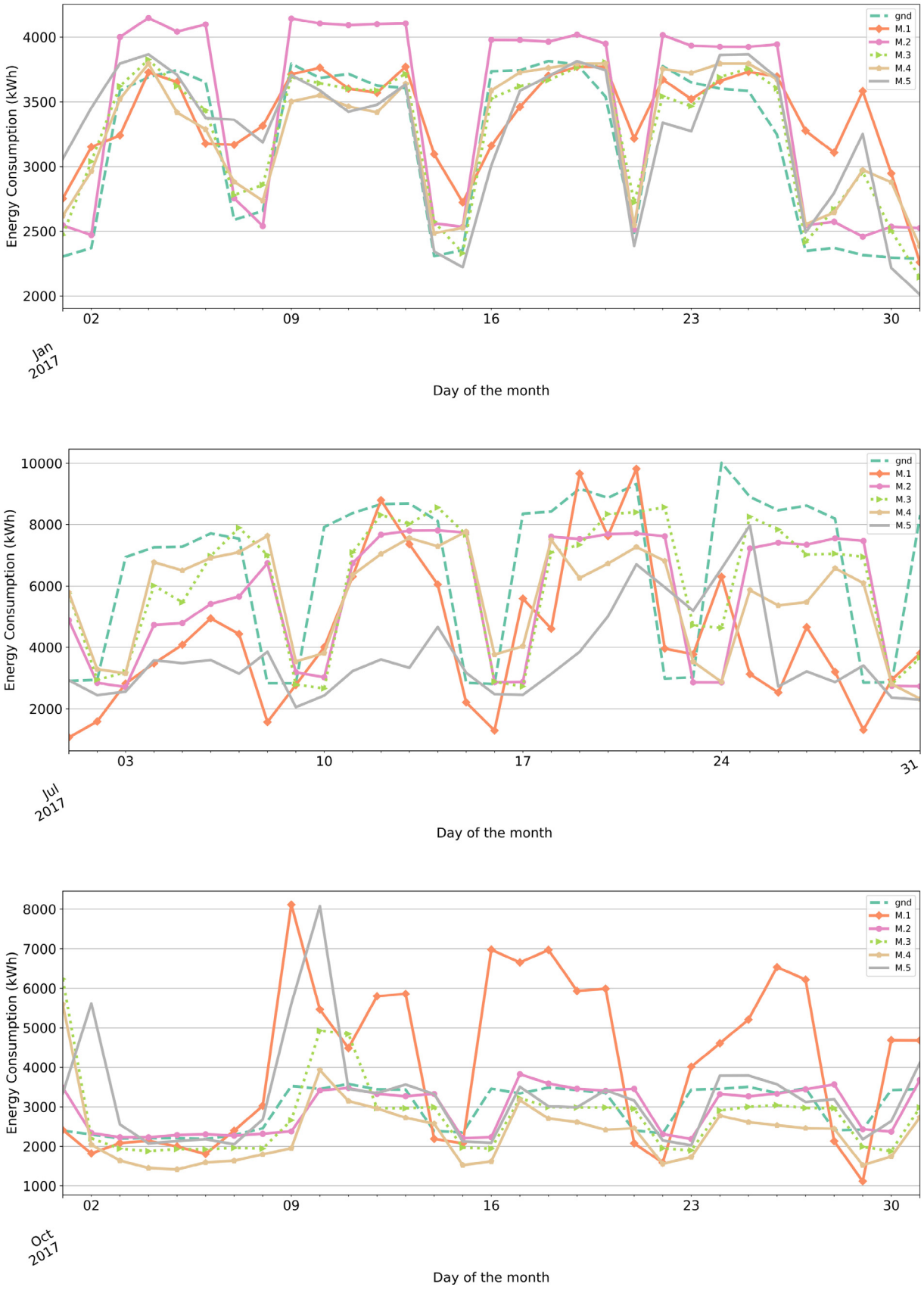


Fig. 16. Three months results of different model in Building A.

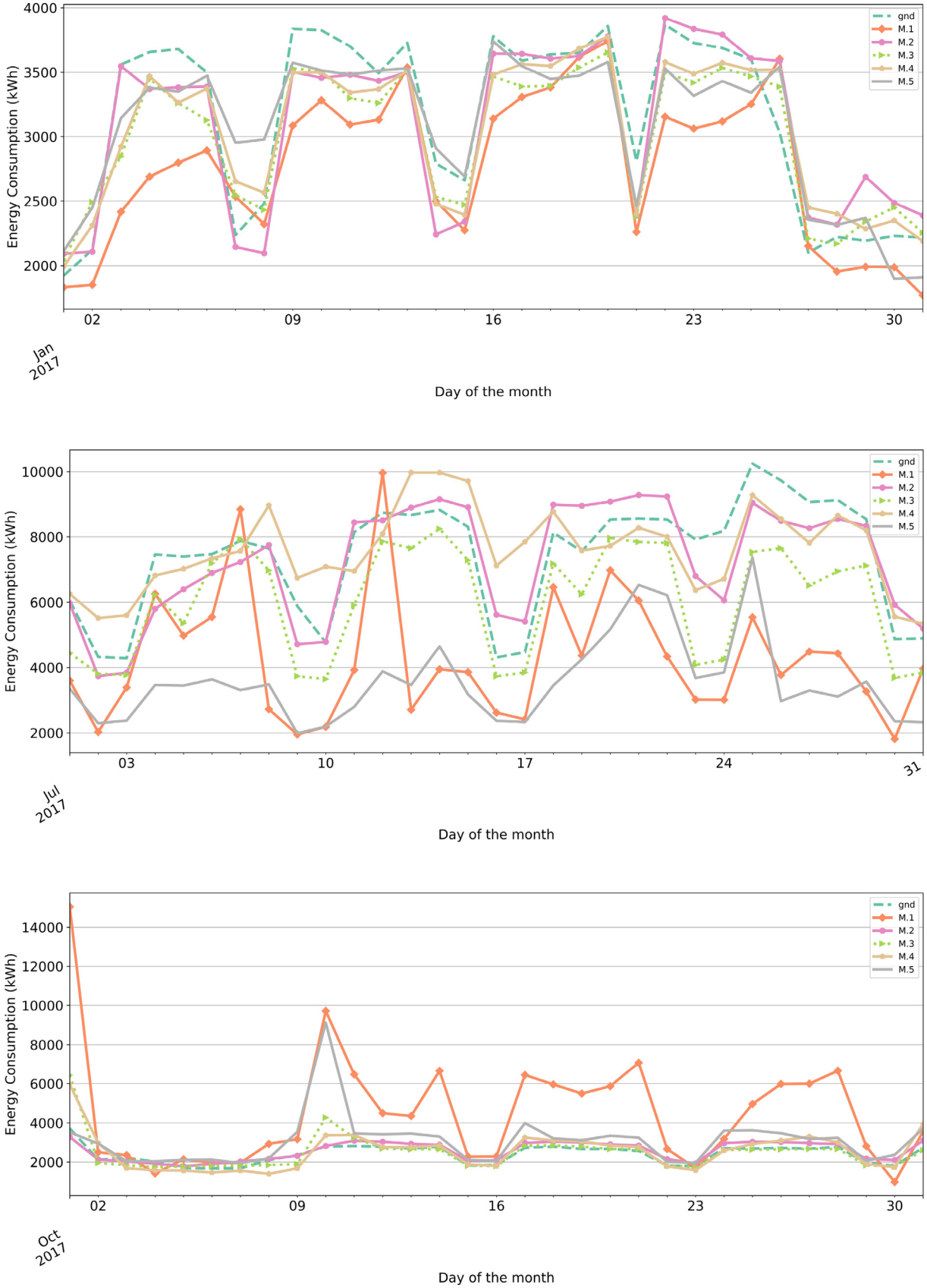


Fig. 17. Three months results of different model in Building B.

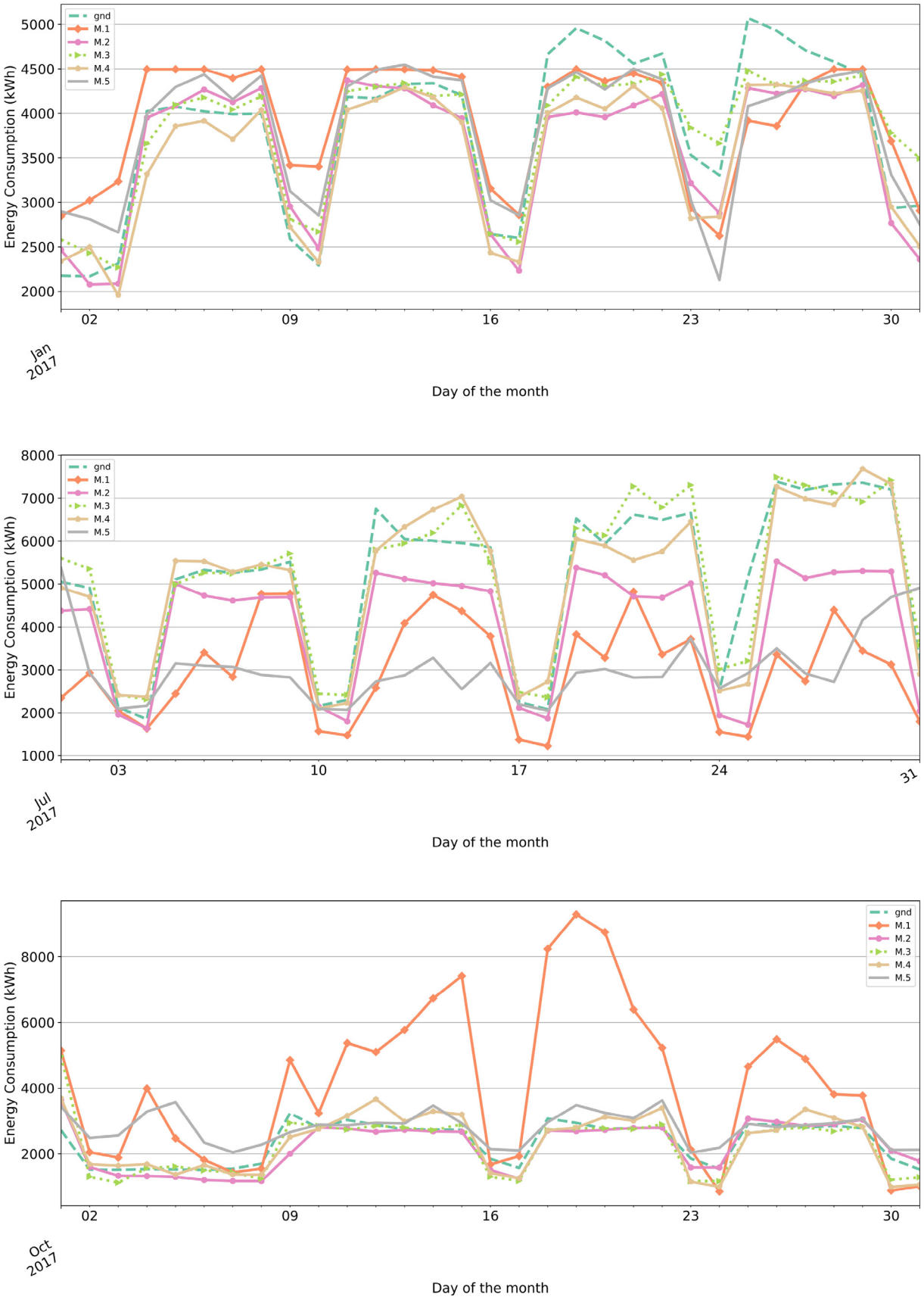


Fig. 18. Three months results of different model in Building C.

Table 8
Use of source domain data in the supplementary experiment.

Supplementary Experiment ID	Source domain
S.1	Data on buildings A for 2015
S.2	Data on buildings A for 2015–2016
S.3	Data on buildings A and C for 2016

Table 9
Evaluation metric results for model predictions in each supplementary experiment.

Supplementary Experiment ID	2D-CNN		Seq2seq	
	MAPE	CV-RMSE	MAPE	CV-RMSE
S.1	0.1417	0.2339	0.1397	0.1960
S.2	0.1324	0.2216	0.1314	0.2344
S.3	0.1205	0.2000	0.1231	0.1928
Original Experiment	0.1187	0.2038	0.1201	0.1707

Compared with the S.1 experiment, the source domain of the S.2 experiment contains 2 years' data of the same building. In the CNN model, MAPE in the S.2 experiment was 0.93 percentage points lower than that in the S.1 experiment, and it was 0.83 percentage points lower in Seq2seq model. However, in the Seq2seq model, the CV-RMSE of the S.2 experiment was 3.84 percentage points higher than that of the S.1 experiment. Therefore, whether adding the data from the same building in the source domain will improve the prediction accuracy needs further research and verification.

S.3 experiment contains two years data from two different buildings in the source domain. Therefore, compared with the S.1 experiment, the source domain of the S.3 experiment has more data, and the similarity degree of the source data in the S.3 experiment is lower than that in the S.2 experiment since the source domain in S.3 contains two buildings' data. The result shows that the prediction of S.3 was better than S.2 and S.1. In 2D-CNN model, the MAPE of S.3 is 2.12 percentage points lower than S.1 and 1.19 lower than S.2. In Seq2seq model, S.3's MAPE is 1.66 percentage points lower than S.1 and 0.83 lower than S.2.

What's more, the results of S.3 are relatively close to the original experiment. In the 2D-CNN model, the difference between MAPE of the two experiments was only 0.18 percentage points, and the CV-RMSE of S.3 is even slightly lower than that of the original experiment. In Seq2seq model, the difference of MAPE between the two models is only 0.3 percentage points.

Therefore, for office buildings, only a complete year's building historical energy consumption data can be used to build the source domain for transfer learning using the model proposed in this paper. However, the buildings in the source domain must share the same functional type and similar climate conditions with the target building. At the same time, the model improves accuracy when the source domain has more data but from different buildings. However, the relationship between the increase of the data volume of the same building in the source domain and the prediction accuracy still needs further study.

Finally, this paper studies three office buildings in Shandong province, China. From the Fig. 11, it can be seen that the farthest distance of the three buildings is 92 km, so the difference in climate conditions is not very large. All three buildings are government office buildings with the same operating schedules. In addition, Ribeiro's study [29] is based on four school buildings in Newfoundland, Canada. The climate condition is not too distinct from one another, and the running schedules of several schools are also the same. The two transfer learning articles are studies of several buildings of the same type with little difference in weather conditions. However, climate condition [52] and operation schedule [9,53] are important factors that determine building energy consumption. Depending on the type of building, the use schedule is often quite different. Thus, if different types of buildings or buildings in different meteorological conditions are selected for source and target buildings, the data of building energy consumption may vary greatly. For example, the use time of office buildings and residential buildings is different, which may lead to different peaks and valleys of energy consumption in a week. The rest day is the peak of energy use of residential buildings, but it is the trough of energy use of office buildings. In this case, if office buildings and residential buildings are used for transfer learning, there may be a big error. Be that as it may, whether these different buildings can apply transfer learning still needs further verification. This part will be taken as the goal of future research.

6. Conclusions

This paper has proposed two novel deep learning models, (1) a 2D CNN with an attention layer and (2) a seq2seq model based on a transfer learning framework. The purpose of these models is to improve the prediction performance for a target building with little high-quality information available by using historical data from similar buildings.

To demonstrate the effectiveness of the models, we have presented a case study on three government office buildings in China. Comparing with the results of model which simulate the poor information situation, the proposed seq2seq model improved MAPE by 19.69 percentage points on average and CV-RMSE by 31.18 percentage points on average; the 2D CNN model improved MAPE by 20.54 percentage points on average and CV-RMSE by 30.32 percentage points on average. Throughout a year of predictions, those two models used one month of data to fine-tune the model pretrained on two years of data (2015 to 2016) from each building in the source domain. The prediction accuracy was close to that using two years of historical data from the target building itself, which demonstrates the effectiveness of these two models. In addition, this study used a model based on simple LSTM layers to evaluate the need to use more complex models; the results demonstrate that although the simple model can hardly improve prediction accuracy over the kind of poor information model, it is worthwhile to apply the more complex proposed models as these can significantly improve the forecasting results considerably over the simple model. Finally, the use and limitation of the model is discussed. Taking building B as the target building, the author designed three different experiments to verify the performance of the transfer learning model under different pre-training data. The results show that the source domain only needs one year's data to achieve a relatively good accuracy. Besides, Using the data from different buildings in the source domain to pretrain can improve the final prediction accuracy to some extent.

Our future research will concentrate on three aspects: (1) For the 2D CNN model, future work will use a 3D convolutional neural network to consider temporal dependency in the process of extracting features from the raw data and (2) the seq2seq model will consider predictions for longer time steps. At the same time, we plan to use the calculation time of the two models as an index for evaluating the quality of the models. (3) The method proposed in this paper is verified on three buildings of the same functional type and climate. Buildings that meet the above conditions have similar total energy profiles, which is beneficial to transfer learning. This paper does not carefully explore the effect of this method in different types of source and target buildings or different climate zones. This content is worthy for future research.

Acknowledgment

This research has been supported by the national key R & D project (No.2016YF0700104-05).

Appendix

Tables 10–12.

Table 10
Data information of M.1.

	Data content	Sampling rate	Function
Train data	The data for 2016–12 on the building A (Features and Labels)	7 days	training
Test data	The data for 2017–1 on the target building (Features)	7 days	Input features at prediction time

Table 11
Data information of M.2.

	Data content	Sampling rate	Function
Train data	The data for 2015 to 2016 on the building A (Features and Labels)	7 days	training
Test data	The data for 2017–1 on the target building (Features)	7 days	Input features at prediction time

Table 12
Data information of M.5.

	Data content	Sampling rate	Function
Source data	The data for 2015 to 2016 on the building B and building C (Features and Labels)	7 days	Pretraining
Target data	The data for 2016–12 on the building A (Features and Labels)	7 days	Fine-tuning
Predictive data	The data for 2017–1 on the target building (Features)	7 days	Final target month forecast

References

[1] J. Eom, L.E. Clarke, S.H. Kim, G.P. Kyle, P.L. Patel, China's building energy use: a long-term perspective based on a detailed assessment, Office Sci. Tech. Inf. Tech. Rep. 199 (1) (2012) 1–14.
[2] Y. Zhang, C.-Q. He, B.-J. Tang, Y.-M. Wei, China's energy consumption in the building sector: a life cycle approach, Energy Build. 94 (2015) 240–251.
[3] B. Yildiz, J.I. Bilbao, A.B. Sproul, A review and analysis of regression and machine learning models on commercial building electricity load forecasting, Renew. Sustain. Energy Rev. 73 (2017) 1104–1122.
[4] J. Steinfeld, A. Bruce, M. Watt, Peak load characteristics of Sydney office buildings and policy recommendations for peak load reduction, Energy Build. 43 (9) (2011) 2179–2187.
[5] W. Li, P. Xu, X. Lu, H. Wang, Z. Pang, Electricity demand response in China: status, feasible market schemes and pilots, Energy 114 (2016) 981–994.
[6] S. Qiu, Z. Li, Z. Pang, W. Zhang, Z. Li, A quick auto-calibration approach based on normative energy models, Energy Build. 172 (2018) 35–46.
[7] Y. Ding, Q. Zhang, T. Yuan, Research on short-term and ultra-short-term cooling load prediction models for office buildings, Energy Build. 154 (2017) 254–267.
[8] S. Heiple, D.J. Sailor, Using building energy simulation and geospatial modeling techniques to determine high resolution building sector energy consumption profiles, Energy Build. 40 (8) (2008) 1426–1436.
[9] Z. Pang et al., Application of mobile positioning occupancy data for building energy simulation: an engineering case study, Build. Environ. 141 (2018) 1–15.
[10] D. Yan, Z. Qiang, T. Yuan, K. Yang, Model input selection for building heating load prediction: a case study for an office building in Tianjin, Energy Build. 159 (2018) 254–270.
[11] S. S. Roy, R. Roy, V. E. Balas, "Estimating heating load in buildings using multivariate adaptive regression splines, extreme learning machine, a hybrid model of MARS and ELM," Renewable & Sustainable Energy Reviews, p. S1364032117308961, 2017.
[12] T. Ahmad, H. Chen, "Short and Medium-term Forecasting of Cooling and Heating load demand in Building Environment with Data-Mining based Approaches," Energy & Buildings, vol. 166, p. S0378778817324659, 2018.
[13] S. Huang, W. Zuo, M.D. Sohn, A Bayesian Network model for predicting cooling load of commercial buildings, Build. Simul. 11 (1) (2018) 87–101.
[14] M.A. Mat Daut, M.Y. Hassan, H. Abdullah, H.A. Rahman, M.P. Abdullah, F. Hussin, Building electrical energy consumption forecasting analysis using conventional and artificial intelligence methods: a review, Renew. Sustain. Energy Rev. 70 (2017) 1108–1118.
[15] M.C. Leung, N.C.F. Tse, L.L. Lai, T.T. Chow, The use of occupancy space electrical power demand in building cooling load prediction, Energy Build. 55 (2012) 151–163.
[16] Z. Pang, Z. O'Neill, Uncertainty quantification and sensitivity analysis of the domestic hot water usage in hotels, Appl. Energy 232 (2018) 424–442.
[17] E. Mocanu, P.H. Nguyen, M. Gibescu, W.L. Kling, Deep learning for estimating building energy consumption ☆, Sustain. Energy Grids Networks 6 (2016) 91–99.
[18] C. Fan, F. Xiao, Y. Zhao, A short-term building cooling load prediction method using deep learning algorithms, Appl. Energy 195 (2017) 222–233.
[19] C. Fan, Y. Sun, Y. Zhao, M. Song, J. Wang, Deep learning-based feature engineering methods for improved building energy prediction, Appl. Energy 240 (2019) 35–45.
[20] C. Fan, J. Wang, W. Gang, S. Li, Assessment of deep recurrent neural network-based strategies for short-term building energy predictions, Appl. Energy 236 (2019) 700–710.
[21] Z. Guo, K. Zhou, X. Zhang, S. Yang, A deep learning model for short-term power load and probability density forecasting, Energy 160 (2018) 1186–1200.
[22] D. Hsu, How much information disclosure of building energy performance is necessary?, Energy Policy 64 (6) (2014) 263–272
[23] A. Farhadi, D.A. Forsyth, R. White, Transfer learning in sign language, IEEE Conference on Computer Vision & Pattern Recognition, 2007.
[24] S. Ruder, "Neural Transfer Learning for Natural Language Processing," p. 329.
[25] X. Cao, D. Wipf, W. Fang, G. Duan, S. Jian, A practical transfer learning algorithm for face verification, IEEE International Conference on Computer Vision, 2013.
[26] J. Gao, H. Ling, W. Hu, J. Xing, Transfer learning based visual tracking with Gaussian processes regression, in: Computer Vision – ECCV 2014, Cham, 2014, pp. 188–203.
[27] T. Grubinger, G.C. Chasparis, T. Natschläger, Generalized online transfer learning for climate control in residential buildings, Energy Build. 139 (2017) 63–71.
[28] E. Mocanu, P.H. Nguyen, W.L. Kling, M. Gibescu, Unsupervised energy prediction in a Smart Grid context using reinforcement cross-building transfer learning, Energy Build. 116 (2016) 646–655.
[29] M. Ribeiro, K. Grolinger, H.F. ElYamany, W.A. Higashino, M.A.M. Capretz, Transfer learning with seasonal and trend adjustment for cross-building energy forecasting, Energy Build. 165 (2018) 352–363.
[30] E. Haselsteiner, G. Pfurtscheller, Using time-dependent neural networks for EEG classification, IEEE Trans. Rehab. Eng. 8 (4) (2000) 457–463.
[31] B. Ganesh, V.V. Kumar, K.Y. Rani, Modeling of batch processes using explicitly time-dependent artificial neural networks, IEEE Trans. Neural Networks Learn. Syst. 25 (5) (2014) 970–979.
[32] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, Nature 521 (2015) 436.
[33] J. Lu, V. Behbood, P. Hao, H. Zuo, S. Xue, G. Zhang, Transfer learning using computational intelligence: a survey, Knowl.-Based Syst. 80 (2015) 14–23.
[34] S.J. Pan, Q. Yang, A survey on transfer learning, IEEE Trans. Knowl. Data Eng. 22 (10) (2010) 1345–1359.
[35] Z. Peng, W. Zhang, N. Han, X. Fang, P. Kang, and L. Teng, "Active Transfer Learning," IEEE Transactions on Circuits and Systems for Video Technology, pp. 1–1, 2019.

[36] D. Wang, T.F. Zheng. "Transfer learning for speech and language processing", in: 2015 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA) Hong Kong, 2015, pp. 1225–1237.
[37] K. Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," arXiv:1406.1078 [cs, stat], Jun. 2014.
[38] I. Sutskever, O. Vinyals, and Q. V. Le, "Sequence to Sequence Learning with Neural Networks," arXiv:1409.3215 [cs], Sep. 2014.
[39] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[40] K. Greff, R.K. Srivastava, J. Koutnik, B.R. Steunebrink, J. Schmidhuber, LSTM: a search space odyssey, IEEE Trans. Neural Networks Learn. Syst. 28 (10) (2017) 2222–2232.
[41] A. Vaswani et al., "Attention Is All You Need," arXiv:1706.03762 [cs], Jun. 2017.
[42] I. Goodfellow, Y. Bengio, A. Courville, Deep Learning, MIT Press, 2016.
[43] A. Baba, S. Adachi, "Simultaneous state of charge and parameter estimation of lithium-ion battery using log-normalized unscented Kalman Filter," in: 2015 American Control Conference (ACC), Chicago, IL, USA, 2015, pp. 311–316.
[44] H. Jaeger, Harnessing nonlinearity: predicting chaotic systems and saving energy in wireless communication, Science 304 (5667) (2004) 78–80.
[45] B. Shi, X. Bai, C. Yao, "An End-to-End Trainable Neural Network for Image-based Sequence Recognition and Its Application to Scene Text Recognition," arXiv:1507.05717 [cs], Jul. 2015.
[46] M. Dwarampudi, N. V. S. Reddy, "Effects of padding on LSTMs and CNNs," arXiv:1903.07288 [cs, stat], Mar. 2019.
[47] A. S. Razavian, H. Azizpour, J. Sullivan, and S. Carlsson, "CNN Features off-the-shelf: an Astounding Baseline for Recognition," arXiv:1403.6382 [cs], Mar. 2014.
[48] L. Perez, J. Wang, "The Effectiveness of Data Augmentation in Image Classification using Deep Learning," arXiv:1712.04621 [cs], Dec. 2017.
[49] M. Peters, S. Ruder, N. A. Smith, "To Tune or Not to Tune? Adapting Pretrained Representations to Diverse Tasks," arXiv:1903.05987 [cs], Mar. 2019.
[50] J. Howard, S. Ruder, "Universal Language Model Fine-tuning for Text Classification," arXiv:1801.06146 [cs, stat], Jan. 2018.
[51] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, R. Salakhutdinov, "Dropout: A Simple Way to Prevent Neural Networks from Overfitting," p. 30.
[52] M. Zhu, Y. Pan, Z. Huang, "The Construction of Minimum Variables Set for Energy Prediction Models of Office Buildings," p. 8, 2019.
[53] G. Jiefan, X. Peng, P. Zhihong, C. Yongbao, J. Ying, C. Zhe, Extracting typical occupancy data of different buildings from mobile positioning data, Energy Build. 180 (2018) 135–145.